Calculus Math Notes

Alexys Ocegueda Castro

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Contents

Chapter 1

Limits

Used in defining the important concepts in calculus such as continunity, derivative of a function, the definite integral of a function.

1.1 What are Limits

It's easy to brush off limits due to their inherent simplicity in solving for them, but it's good to know what is happening when you say a limit equals something. This means that as you zoom in on that one area it is *approaching*, the number it equals, despite it potentially never reaching it, still is the answer. The limit is the number you inch closer and closer to while zooming in.

1.1.1 Inituitive Definition

The **limit** of a function describes the behavior of a function near a particular value of x.

When a limit heads towards infinity due to a vertical asymptote this is said to be a **infinite limit**.

1.2 One-Sided Limits

When a limit has a '-' sign that means the limit as it approaches from the *left*.
When a limit has a '+' sign that means the limit as it approaches from

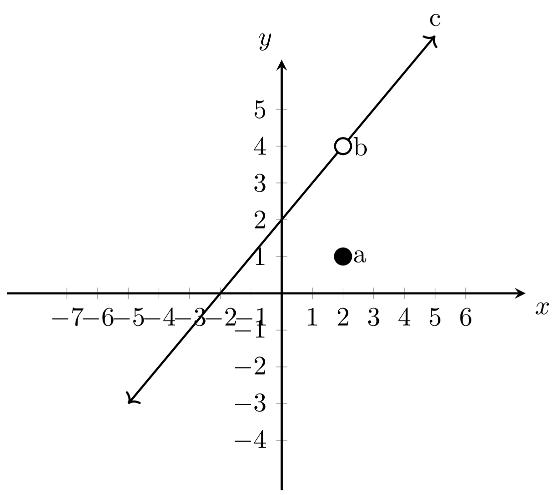
the *right*.

1.3 Trigonometric Functions

There are four important limit functions:
1. $\lim_{x \to c} sinx = sinc$
2. $\lim_{x \to c} cosx = cosc$
3. $\lim_{x \to 0} \frac{sinx}{x} = 1$
4. $\lim_{x \to 0} \frac{1-cosx}{x} = 0$

1.4 Continunity

Three conditions must be met for a point to considered continuous:
1. f(c) exists
2. $\lim_{x \to c} f(x)$
3. $\lim_{x \to c} f(x) = f(c)$



In the graph above we can say multiple things:

1. A limit exists at $f(b)$ which is 4.
2. The limit *at* $f(b)$ is 1 aka 'a'.
3. This line has a disconnected continunity.

A test would be to imagine a pencil going along $f(x)$ without having to be picked up – if it can accomplish this then it is continuous.

Chapter 2

Derivatives

2.1 What is a Derivative

It is the instantaneous rate of change which says how much something changes in relation to another. It's the slope of the tangent line at any given point revealing steepness and direction (increasing/decreasing).

2.1.1 Definition

The derivative of a function $y = f(x)$ at point $(x, f(x))$, is defined as:

$$\lim_{\delta x \to 0} \tfrac{f(x+\delta x)-f(x)}{\delta x}$$

Example: Using the definition, find the derivative of $f(x) = 5x - 4$

You will want to add in delta into $f(x)$ when you utilize the definition.

$$\lim_{\delta x \to 0} \frac{(5(x+\delta x)-4)-(5x-4)}{\delta x}$$

$$\lim_{\delta x \to 0} \frac{5x+5\delta x-4-5x+4)}{\delta x}$$

$$\lim_{\delta x \to 0} \frac{5\delta x}{\delta x}$$

$$\lim_{\delta x \to 0} \frac{5}{1}$$

$$5$$

Note that **Differentiability implies continuity, but continuity *does not* imply differentiability**

2.2 Differentiation Rules

1. If $f(x) = c$ where c is a constant, then $f'(x) = 0$.
2. If $f(x) = c * g(x)$, then $f'(x) = c \cdot g'(x)$.
3. Sum Rule: If $f(x) = g(x) + h(x)$ then $f'(x) = g'(x) + h'(x)$.
4. Difference Rule: If $f(x) = g(x) - h(x)$, then $f'(x) = g'(x) - h'(x)$.
5. Product Rule: If $f(x) = g(x) \cdot h(x)$, then $f'(x) = f(x)g'(x) + g(x)f'(x)$.
6. Quotient Rule: If $f(x) = g(x)/h(x)$, then $f'(x) = \frac{g'(x) \cdot h(x) - h'(x)g(x)}{[h(x)]^2}$

2.3 Trigonometric Functions

The 6 trigonometric functions also have rules:
1. If $f(x) = sinx$, then $f'(x) = cosx$
2. If $f(x) = cosx$, then $f'(x) = -sinx$
3. If $f(x) = tanx$, then $f'(x) = sec^2x$
4. If $f(x) = cotx$, then $f'(x) = -csc^2x$
5. If $f(x) = secx$, then $f'(x) = secxtanx$
6. If $f(x) = cscx$, then $f'(x) = -cscxcotx$

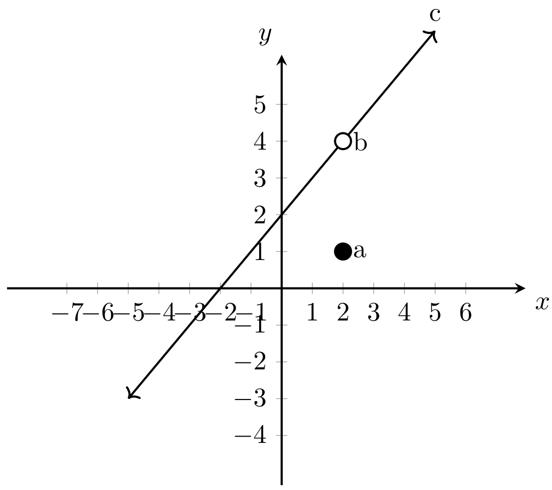
2.4 Chain Rule

The **chain rule** is a technique for finding the derivative of composite functions. With the number of differentiation steps dependent on how many functions there are.

Example 1:
$f(x) = (g \circ h)(x) = g[h(x)]$ then
$f'(x) = g'[h(x)] \cdot h'(x)$

Example 2:
$r(x) = (m \circ n \circ p)(x) = m\{n[p(x)]\}$
then $r'(x) = m'\{n[p(x)]\} \cdot n'[p(x)] \cdot p'(x)$

2.5 Implicit Differentiation



c
y
b
a
x